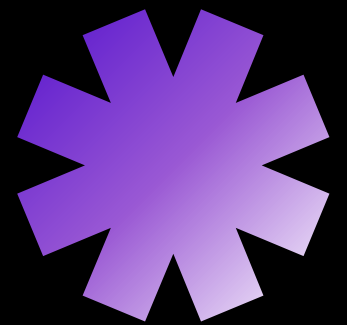
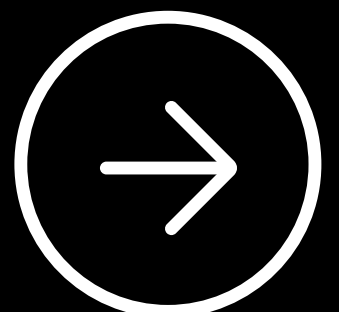




# DEVOPS PRACTICE

## DAY 14 – DAY 15

- Networking fundamentals
- DNS and IP concepts
- CIDR and subnetting
- Ports and services
- Linux networking troubleshooting
- AWS EC2 networking practice

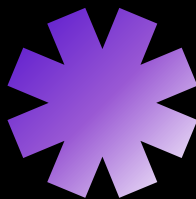


## Networking Basics

# 14

- OSI vs TCP/IP model
- IP address basics
- Public vs private IP
- DNS resolution
- TCP and UDP basics
- HTTP and HTTPS concepts

```
ubuntu@ip-172-31-3-219:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 06:f6:f5:89:1f:2f brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx06f6f5891f2f
    inet 172.31.3.219/20 metric 100 brd 172.31.15.255 scope global dynamic ens5
        valid_lft 2563sec preferred_lft 2563sec
    inet6 fe80::4f6:f5ff:fe89:1f2f/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```



Troubleshooting

Commands

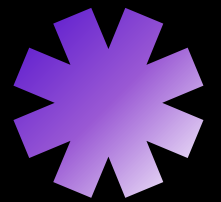
15

- ping command
- traceroute practice
- curl HTTP checks
- ss and netstat
- nc port testing
- Linux networking troubleshooting

```

ubuntu@ip-172-31-3-219:~$ traceroute google.com
traceroute to google.com (142.250.69.78), 30 hops max, 60 byte packets
 1  240.4.160.15 (240.4.160.15)  1.369 ms 240.4.160.13 (240.4.160.13)  1.372 ms 240.4.160.12 (240.4.160.12)  1.594 ms
 2  242.12.188.131 (242.12.188.131)  1.317 ms 242.12.188.1 (242.12.188.1)  1.327 ms 242.12.188.129 (242.12.188.129)  1.311 ms
 3  99.83.65.228 (99.83.65.228)  1.379 ms 1.866 ms 99.83.65.230 (99.83.65.230)  1.399 ms
 4  99.83.65.229 (99.83.65.229)  0.917 ms 99.83.65.231 (99.83.65.231)  0.834 ms 99.83.65.229 (99.83.65.229)  0.894 ms
 5  * 192.178.86.185 (192.178.86.185)  1.767 ms *
 6  142.250.58.131 (142.250.58.131)  2.047 ms 108.170.232.67 (108.170.232.67)  1.855 ms 1.784 ms
 7  tzyula-aa-in-f14.1e100.net (142.250.69.78)  1.016 ms 1.048 ms 1.027 ms
ubuntu@ip-172-31-3-219:~$ ss -tulpn
Netid  State  Recv-Q  Send-Q      Local Address:Port      Peer Address:Port  Process
udp    UNCONN  0        0      127.0.0.54:53           0.0.0.0:*
udp    UNCONN  0        0      127.0.0.53%lo:53       0.0.0.0:*
udp    UNCONN  0        0  172.31.3.219%ens5:68    0.0.0.0:*
udp    UNCONN  0        0      127.0.0.1:323          0.0.0.0:*
udp    UNCONN  0        0      [::1]:323             [::]:*
tcp    LISTEN  0       4096      0.0.0.0:22             0.0.0.0:*
tcp    LISTEN  0       4096  127.0.0.53%lo:53       0.0.0.0:*
tcp    LISTEN  0       4096      127.0.0.54:53         0.0.0.0:*
tcp    LISTEN  0       4096      [::]:22               [::]:*
ubuntu@ip-172-31-3-219:~$

```



# 13

## CIDR & Ports

### Networking Concepts

- /24, /16, /28 subnet understanding
- Port 22 → SSH
- Port 80 → HTTP
- Port 443 → HTTPS
- Port 53 → DNS
- Port 3306 → MySQL

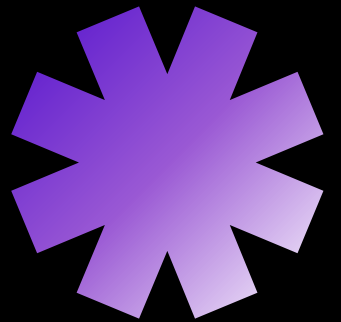
```
ubuntu@ip-172-31-3-219:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53
```

```
Non-authoritative answer:
Name:   google.com
Address: 142.250.69.46
Name:   google.com
Address: 2607:f8b0:4020:c07::66
Name:   google.com
Address: 2607:f8b0:4020:c07::65
Name:   google.com
Address: 2607:f8b0:4020:c07::8b
Name:   google.com
Address: 2607:f8b0:4020:c07::8a
```

```
ubuntu@ip-172-31-3-219:~$
```



# Consistency Daily Practice Hands-On Learning



- DNS and IP understanding
- Networking troubleshooting
- Linux networking tools
- AWS EC2 practice
- Real DevOps networking basics

```
ubuntu@ip-172-31-3-219:~$ netstat -an | head
Command 'netstat' not found, but can be installed with:
sudo apt install net-tools
ubuntu@ip-172-31-3-219:~$ sudo apt install net-tools
Installing:
  net-tools

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 41
  Download size: 152 kB
  Space needed: 680 kB / 17.2 GB available

Get:1 http://ca-central-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 net-tools amd64 2.
0-2ubuntu1 [152 kB]
Fetched 152 kB in 0s (10.3 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 85321 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Setting up net-tools (2.10-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-3-219:~$ netstat -an | head
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:53              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:54              0.0.0.0:*               LISTEN
tcp        0      0 172.31.3.219:22        76.69.67.47:58692      ESTABLISHED
tcp        0      0 172.31.3.219:37966     3.96.211.9:80          TIME_WAIT
tcp6       0      0 :::22                  :::*                    LISTEN
udp        0      0 0.0.0.0:53              0.0.0.0:*               LISTEN
udp        0      0 0.0.0.0:54              0.0.0.0:*               LISTEN
ubuntu@ip-172-31-3-219:~$
```